

**To accelerate the world's
transition to sustainable energy**



TESLA



Shaping the EV Future: Economic and Political Forces Driving Change in Tesla's Future



SOCIETAL

- Sustainability is now tied to personal identity and social status, especially among younger generations.
- Urban policies like low-emission zones and car-free areas are reshaping transportation toward shared mobility and EVs.
- EVs represent 16.7% of global sales in 2025, leaving over 80% of the market still untapped.



TECHNOLOGICAL

- The conversation is moving from battery production to battery lifecycle management, where advanced recycling technologies aim to reduce reliance on raw material mining.
- Autonomous vehicle technology faces hurdles in areas with poor infrastructure, inconsistent mapping, and regulatory barriers.
- EVs are becoming active participants in energy grids, supplying power during demand peaks.



ECONOMICAL

- **The global EV market is expected to grow significantly, with a projected market size of over \$1.5 trillion by 2030.**
- **Regional supply chains and resource pooling are addressing raw material cost spikes.**
- **While EV costs are dropping globally, affordability remains a barrier in developing economies where EVs still compete with cheaper, conventional vehicles.**



ENVIRONMENTAL

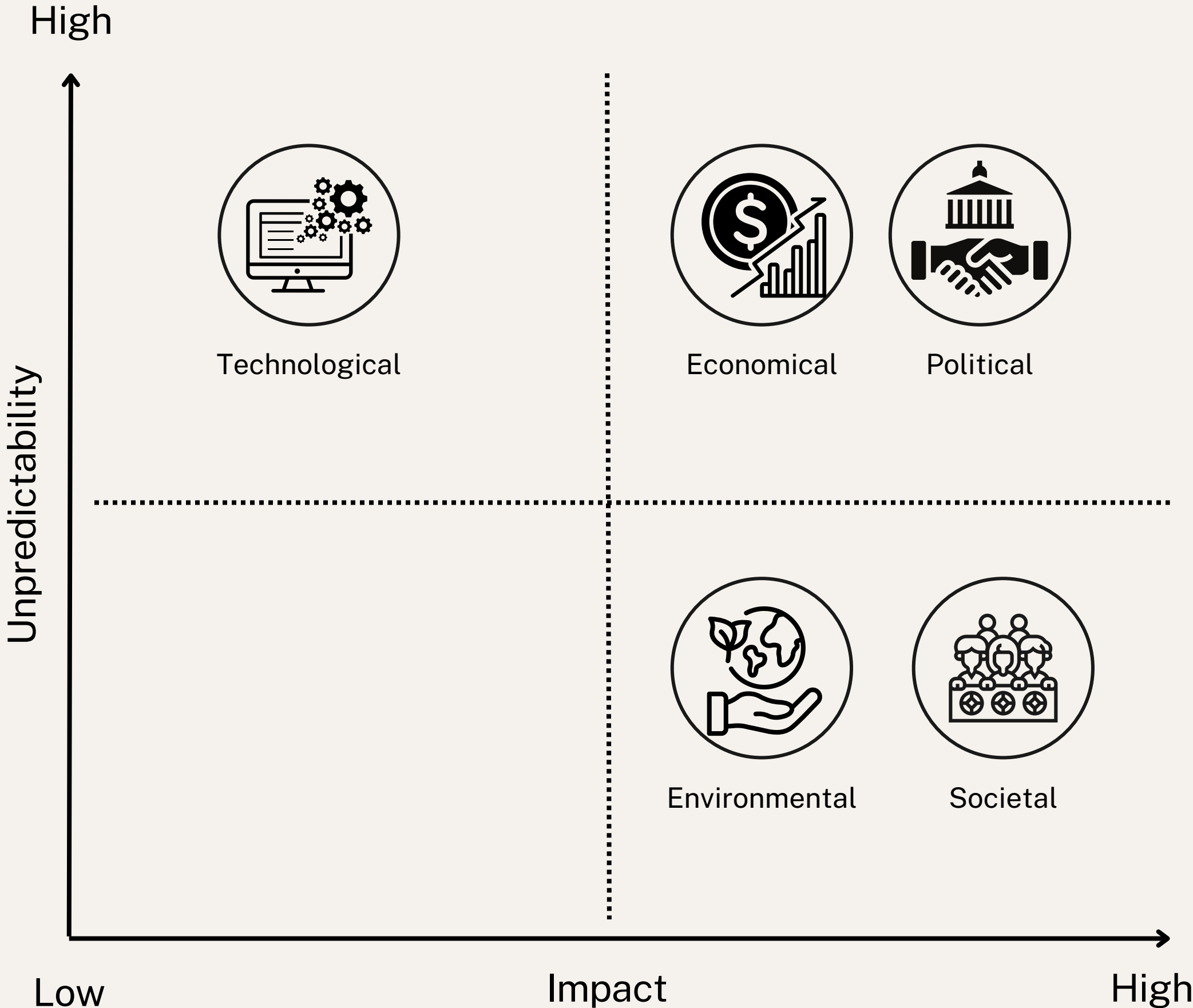
- Uneven recycling infrastructure is a challenge for regions adopting EVs without proper disposal systems.
- Battery designs like solid-state or LFP are reducing reliance on environmentally damaging materials.
- The global shift towards electric vehicles, driven by the Paris Climate Agreement and other climate initiatives, aims to reduce carbon emissions by 2050.



POLITICAL

- **Carbon taxes are pushing automakers to adopt cleaner production processes to remain competitive.**
- **Cross-border alliances are reducing trade barriers and harmonizing EV standards to accelerate adoption.**
- **Geopolitical risks, including U.S.-China tensions and potential government actions, could disrupt supply chain, increase production costs, and impact market access and pricing globally.**

Geopolitical Dynamics and Material Constraints Posing Major Strategic Risks



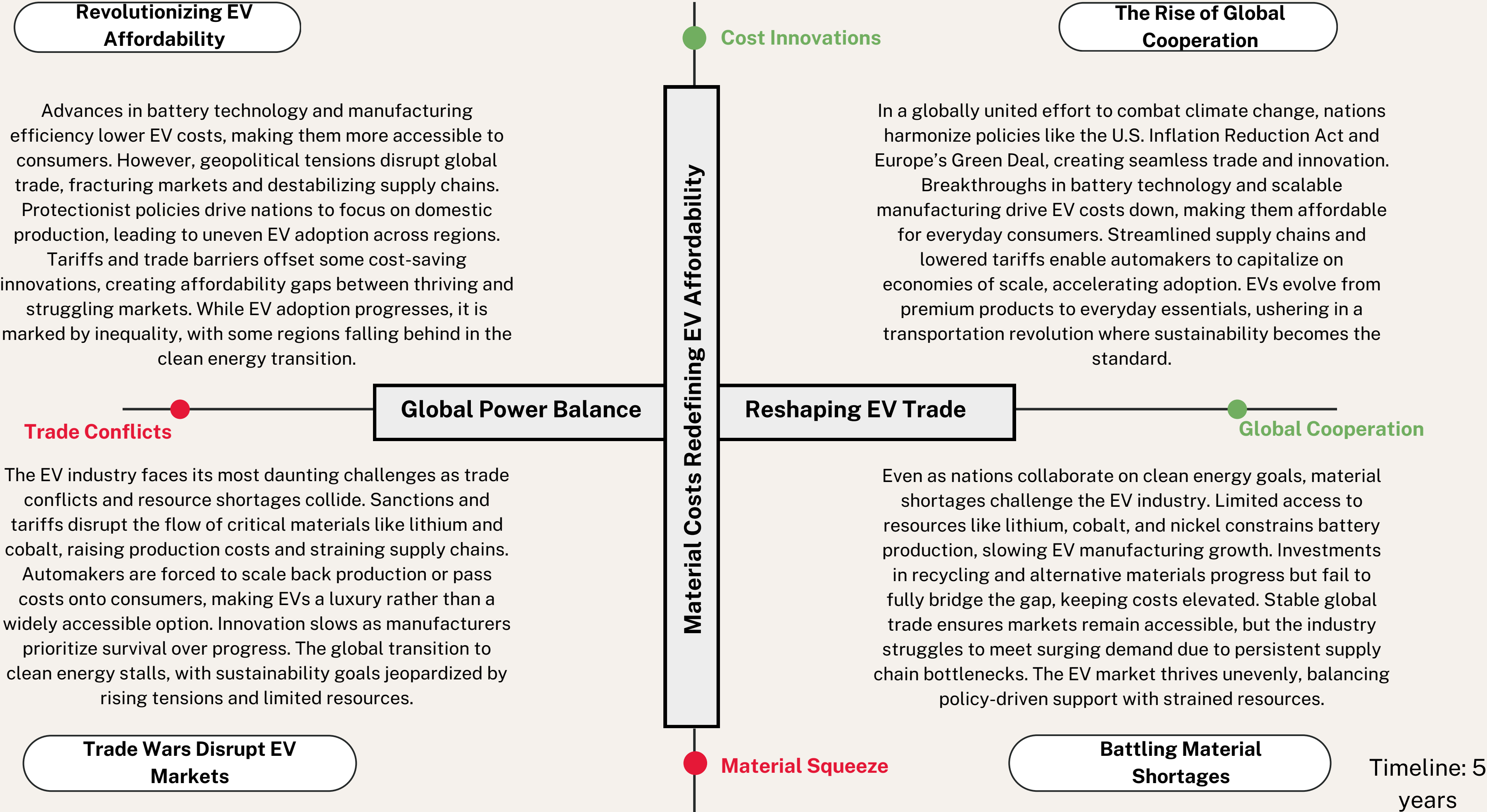
Global Politics Shaping the Future of EV Supply Chains – Political

- **Global Cooperation:** Stable diplomatic relationships and collaborative trade agreements ease tariffs and facilitate the smooth flow of raw materials enabling the EV market to grow unhindered.
- **Trade Conflicts:** Geopolitical tensions, especially between the U.S. and China, could disrupt the global EV supply chain, increase costs, and limit market access.

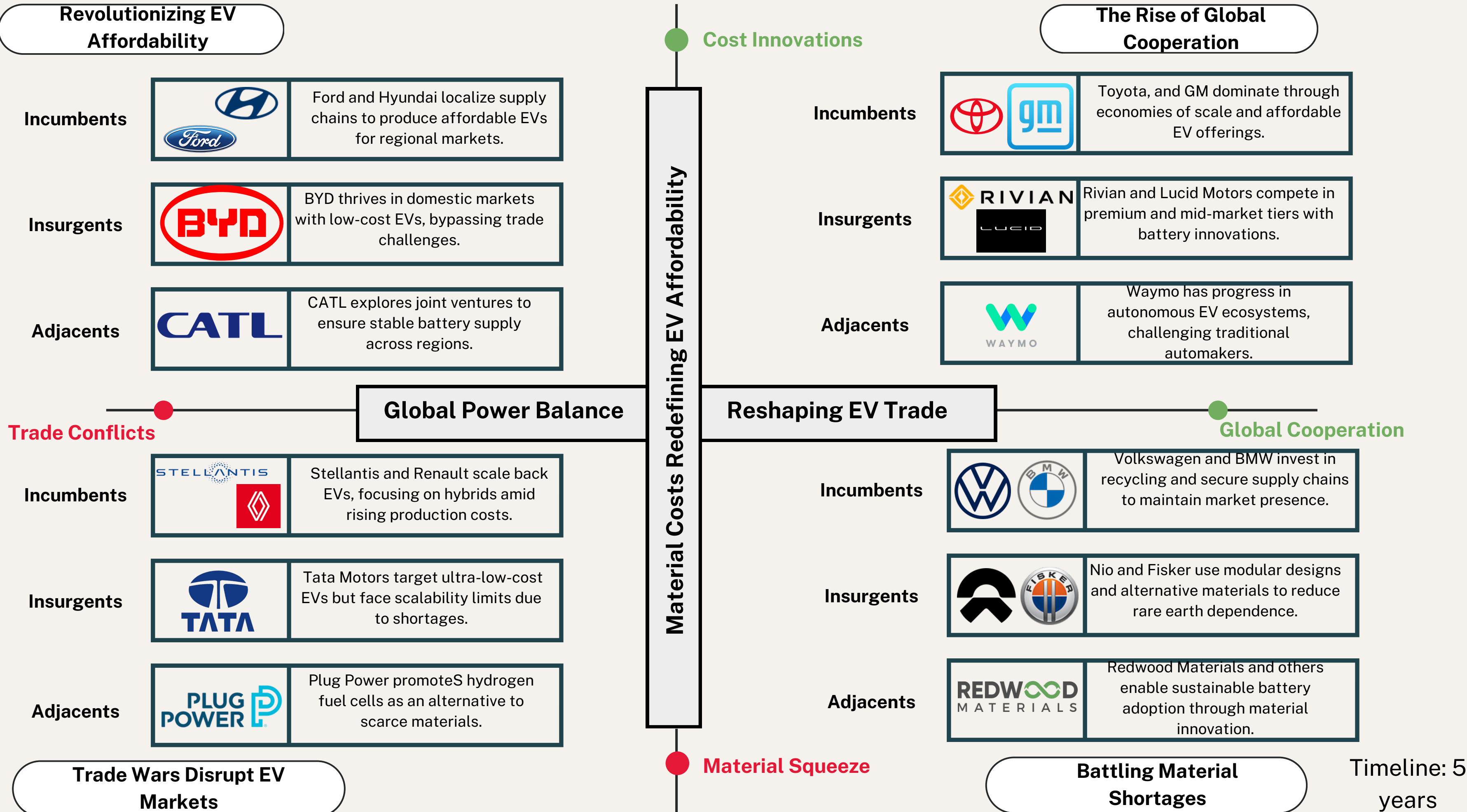
Economic Pressures Transforming EV Affordability and Growth – Economic

- **Cost Innovations:** Advances in battery technology, efficient recycling systems, and alternative materials lower production costs, making EVs more affordable and accelerating adoption across global markets.
- **Material Squeeze:** Limited availability of raw materials like lithium and cobalt, coupled with rising extraction costs, leads to higher EV prices, slowing adoption in price-sensitive regions and creating affordability challenges.

Scenarios for Tesla's Path Forward in a Changing Global Market



Tesla's Position Across Incumbents, Insurgents, and Adjacent Players



Tesla Must Focus on Innovation, Sustainable Practices, and Strategic Responses to Capitalize on Opportunities

DRIVERS

- The future of EV supply chains hinges on geopolitical dynamics. Stable diplomatic relations can streamline material flow and reduce costs, while trade tensions could disrupt Tesla's production and inflate prices, challenging its global market position.
- Breakthroughs in battery technology and recycling could make EVs affordable for mass adoption. Conversely, scarcity of critical raw materials may escalate production costs, limiting Tesla's ability to compete in price-sensitive markets.

OPPORTUNITIES

- Tesla's cutting-edge 4680 battery cells offer higher energy density and cost-efficiency, unlocking new vehicle categories (e.g., trucks, vans) and improving range and performance, positioning Tesla ahead of competitors.
- EVs acting as energy storage units during peak demand provide Tesla with opportunities to integrate seamlessly into renewable energy ecosystems, creating new revenue streams and increasing customer value.
- Harmonization of climate policies (e.g., U.S. Inflation Reduction Act) reduces trade barriers, facilitating Tesla's growth in international markets and enhancing economies of scale.

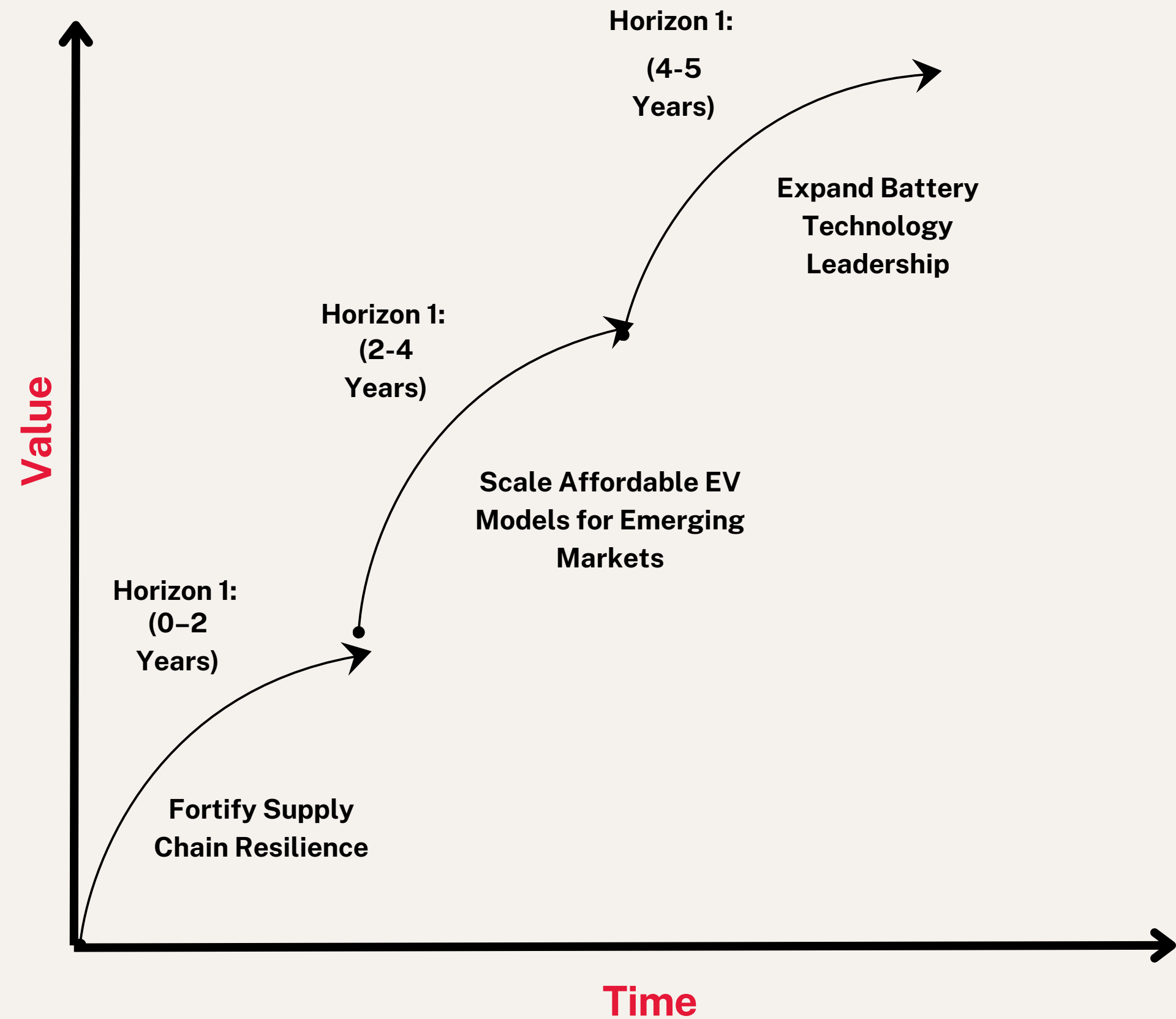
THREATS

- Rising U.S.-China tensions may hinder Tesla's access to raw materials like lithium and cobalt, inflating production costs and slowing EV adoption globally.
- Despite falling EV costs, high price points and limited infrastructure in developing economies could slow adoption, leaving Tesla vulnerable to price-sensitive competitors.
- Uneven recycling infrastructure and reliance on environmentally damaging raw materials expose Tesla to sustainability risks, regulatory scrutiny, and reputational damage.

STRATEGIC RESPONSES

- **Fortify Supply Chain Resilience**
- **Scale Affordable EV Models for Emerging Markets**
- **Expand Battery Technology Leadership**

The Next Five Years Will Require Tesla to Fortify Supply Chains, Scale Models, and Lead Battery Innovation



1. Fortify Supply Chain Resilience

By securing diverse and localized raw material sources, Tesla can shield itself from geopolitical risks and ensure uninterrupted production, maintaining its competitive edge in a rapidly evolving global market.

2. Scale Affordable EV Models for Emerging Markets

Launching cost-effective EVs designed for emerging economies will open Tesla to massive untapped markets, driving widespread EV adoption while solidifying its position as a truly global brand.

3. Expand Battery Technology Leadership

By advancing cutting-edge battery innovation and sustainability efforts, Tesla will redefine energy storage capabilities, making EVs more accessible, efficient, and integrated into renewable energy ecosystems.

Appendix

Executive Summary

- Shaping the EV Future: Economic and Political Forces Driving Change in Tesla's Future
- Geopolitical Dynamics and Material Constraints Posing Major Strategic Risks
- Scenarios for Tesla's Path Forward in a Changing Global Market
- Tesla's Position Across Incumbents, Insurgents, and Adjacent Players
- Tesla Must Focus on Innovation, Sustainable Practices, and Strategic Responses to Capitalize on Opportunities
- The Next Five Years Will Require Tesla to Fortify Supply Chains, Scale Models, and Lead Battery Innovation